Exercises - chapter 2: Integration and Administration

Exercise 1: Conceptual Understanding & Terminology

1- Define the acronym **ETL** and briefly explain the purpose of each step in the context of a Data Warehouse.
2- List and explain three distinct techniques used for **Data Extraction** as mentioned in the course
3- Explain the difference between **Full Loading** and **Incremental Loading**. When would you strictly *must* use Full Loading?
4- In the context of Data Transformation, explain **"Binning"** and provide an example of how it is used.
5- The lecture discusses **"Information Loss"** during transformation.
    a. Identify two specific transformation processes that cause information loss.
    b. Why might a Business Intelligence engineer *intentionally* choose to lose information (e.g., via Aggregation)?
6- The course material contrasts different loading plans. Explain the difference between Serial Loading and Parallel Loading. How does the technique of File Partitioning contribute to the efficiency of Parallel Loading?

Exercise 2: True or False?

Determine if the following statements are True or False. If False, correct the statement.

1. The default loading plan in a Data Warehouse is **Parallel Loading**, where data is added simultaneously to speed up the process.
2. **Web Scraping** is a technique used to connect automatically to data sources via a programmatic interface to query data.
3. **Incremental Loading** is a strategy where the system deletes existing records and reloads the entire dataset from scratch every time.
4. In the **Pull Method**, the Data Warehouse subscribes to receive data, making it the best choice for real-time streaming.

Exercise 3: The "Smart City" Case Study

Scenario: Imagine you are the Data Architect for a city. You need to build a Data Warehouse to analyze traffic and weather patterns. You have two main data sources:

1. **Old Traffic Logs:** 50 years of paper documents containing handwritten records of car accidents.
2. **Live IoT Sensors:** Thousands of sensors on traffic lights measuring car speed and temperature in real-time.

Questions:

- For the **Old Traffic Logs**, which specific extraction technique would you use to digitize the text?

- The city wants to protect the privacy of drivers involved in accidents mentioned in the old logs. Which transformation technique should be applied to names and license plate numbers?
- You need to load the **Live IoT Sensor** data.
    - Would you use **Push** or **Pull** methodology? Explain why.
    - Would you use **Batch Loading** or **Stream Loading**? Explain why.
- You have millions of records coming in from the sensors every hour. To speed up the process, you decide not to load the data one by one (Serial Loading).
    - What loading plan should you adopt to optimize speed?
    - The sensor system aggregates all records into **one single massive file** (50GB) every hour. Your junior engineer argues that Parallel Loading is impossible because there is only "one source file." Explain how you can still apply Parallel Loading to this single file to prove them wrong.